

GANESH ARIVOLI

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SUMMARY

Graduate student in Mechanical Engineering focused on developing GPU-accelerated physics engines for robotics and autonomous systems. Experienced in building high-fidelity simulation frameworks for accurate and fast simulation of complex systems. Skilled in C++, CUDA, and Python (Warp, Taichi) with strong foundations in multibody dynamics, numerical methods, and high-performance computing. Active contributor to [Project Chrono](#), developing GPU-based FEA and linear solver frameworks for large-scale flexible multibody dynamics. Currently working on hydroelastic contact and solver modules for [NVIDIA Newton](#).

EDUCATION

University of Wisconsin-Madison

Master of Science, Mechanical Engineering, Advisor: Dan Negrut

Dec 2026 (Expected)

Current GPA: 4.0/4.0

National Institute of Technology, Tiruchirappalli

Bachelor of Technology, Mechanical Engineering

RELEVANT COURSEWORK

High Performance Computing; Advanced Computational Dynamics; Kinematics, Dynamics and Control of Robotic Manipulators; Feedback Control; Mechatronics; Vehicle Dynamics; Linear Optimization; Computer Vision; Machine Learning

RESEARCH EXPERIENCE

Robotics Intern - Dynamics and Simulation | Toyota Research Institute, CA

May 2026 – Present

- Characterizing hydroelastic contact methods across surface accuracy, geometric generality, robustness, and performance.
- Designing a novel contact representation to improve accuracy and efficiency beyond tetrahedral mesh and SDF-based methods.
- Studied the effect of contact reduction and solver formulation on hydroelastic contact fidelity and efficiency.
- Developing hydroelastic contact and solver modules for [NVIDIA Newton](#)'s GPU-resident physics pipeline.

GPU-Based Total-Lagrangian FEA Framework for Multibody Dynamics | UW-Madison

May 2025 – May 2026

- Developed a GPU-optimized unified Total Lagrangian FEA formulation for flexible multibody dynamics with large deformations, kinematic constraints, hyperelastic and viscoelastic materials, and frictional contact.
- Implemented and optimized a GPU-resident nonlinear solve pipeline with implicit (second-order Newton, first-order AdamW) and explicit (leapfrog) solvers, utilizing BVH-free collision detection.
- Validated against closed-form analytical solutions and achieved a **10× speedup** over CPU baselines at up to 1.7M DOFs.
- Contributed to [open-source implementation](#) and two papers ([Part I](#), [Part II](#)) currently under review.

GPU-Based Linear Solvers for Multibody Dynamics | UW-Madison

Dec 2024 – May 2025

- Investigated scaling of GPU-based sparse direct solver phases (symbolic analysis, factorization, and solve) with problem size, evaluating solve time and solution accuracy for multibody dynamics applications.
- Observed a **3.6× speedup** with CuDSS over Pardiso on large ANCF systems, within engineering accuracy tolerances.
- Results summarized in a [published extended abstract](#) at ASME IDETC-CIE 2025 with [reproducible code](#).

Autonomous Terrain Leveling | UW-Madison

Jan 2025 – May 2025

- Built a CNN-based differentiable surrogate of bulldozer-soil interaction trained on Chrono::FSI physics-based simulation data, enabling gradient-based blade control optimization for autonomous terrain leveling.
- Developed training and inference pipelines on HPC clusters to support large-scale data generation and model deployment.
- Achieved a **48% improvement** in terrain-leveling performance; published in *Advanced Robotics Research* ([DOI](#)).

DEM Simulation for Komatsu Loader | Komatsu Mining Corp, Longview, TX

Jan 2025 – June 2025

- Developed a multibody dynamics model of a Komatsu loader and integrated it with a GPU-accelerated [DEM engine](#), enabling high-fidelity simulation of loader-granular material interaction.
- Conducted simulation-driven parametric analysis of bucket geometry and motion to optimize excavation efficiency.

SELECTED PROJECTS

Physics Engine for Rigid/Flexible Multibody and Robotic Systems | UW-Madison

Aug 2025 – Dec 2025

- Developed a high-fidelity multibody dynamics simulator for rigid and flexible systems using generalized coordinates, constraints, BDF time-stepping, and a custom Newton-Raphson solver.
- Extended the engine to support multi-DOF serial robotic manipulators with kinematics, dynamics, control, trajectory generation, and real-time visualization.

GPU Accelerated Path Planning for Autonomous Robots | UW-Madison

Jan 2025 – Apr 2025

- Implemented parallelized RRT and RRT* algorithms in CUDA (GPU) and OpenMP (multicore CPU), optimizing collision checking and nearest-neighbor routines for high-dimensional planning in obstacle-dense environments ([code](#)).
- Built a benchmarking suite with physics-based simulation to evaluate planning latency, control feasibility, and trajectory smoothness.

ADDITIONAL EXPERIENCE

Vehicle Dynamics Engineer (R&D) | Bajaj Auto, India

Sep 2021 – Dec 2023

- Led the design and development of suspension and steering systems for Bajaj Auto-Triumph motorcycles, integrating physics-based simulation models into the product development cycle and contributing to **100,000+** units sold across **58+** countries.
- Improved **ride and handling performance by 15%** through multibody dynamics simulations, optimizing suspension parameters.
- Key contributor to the [Triumph Speed 400](#) and [Scrambler 400 X](#), launched globally in 2023.

Graduate Teaching Assistant | UW-Madison

Jan 2026 – May 2026

- Supported CS 759, a graduate-level High-Performance Computing programming course covering CUDA, OpenMP, and MPI.
- Held office hours and graded assignments on debugging, performance analysis, and parallel programming projects.

SAE BAJA Racing Team | NITT, India

May 2018 – May 2021

- Led powertrain and data acquisition sub-teams in design, fabrication, and testing of an All-Terrain Vehicle for SAE BAJA India, an intercollegiate off-road engineering competition, securing **consecutive 5th place** finishes in 2020 and 2021 among **100+** teams.
- Built a MATLAB-based CVT model for simulation-driven calibration, improving drivetrain efficiency by **10%** and reducing tuning time by **40%**.

SKILLS

Programming: C/C++, CUDA, Python (Warp, Taichi), OpenMP, MATLAB

Tools & Platforms: Linux, Git, NVIDIA Nsight, Docker, ROS (basic), VSCode, Bazel

Simulation: Project Chrono, Drake, NVIDIA Newton, Mujoco, Simulink, MSC Adams

PUBLICATIONS

- **Ganesh Arivoli**, Huzaifa Unjhwala, Michael Taylor, Radu Serban, Dan Negrut, "On the Use of GPU-Based Linear Solvers in Multibody Dynamics," *Extended abstract presented at the ASME IDETC-CIE 2025*. <https://gnsh-a.github.io/assets/pdf/arivoliz025gpu.pdf>
- Zhenhao Zhou, **Ganesh Arivoli**, Dan Negrut, "A Total Lagrangian Finite Element Framework for Multibody Dynamics: Part I – Formulation," *accepted for publication in Engineering with Computers* (2026). <https://arxiv.org/abs/2602.17002>
- Zhenhao Zhou*, Ruochun Zhang*, **Ganesh Arivoli**, Dan Negrut, "A Total Lagrangian Finite Element Framework for Multibody Dynamics: Part II – GPU Implementation," *under review at Engineering with Computers* (2026). <https://arxiv.org/abs/2604.10357>
- Harry Zhang*, **Ganesh Arivoli***, Huzaifa Unjhwala, Luning Bakke, Radu Serban, Dan Negrut, "Data-Driven Bulldozer Blade Control for Autonomous Terrain Leveling," *Advanced Robotics Research* (2026). <https://doi.org/10.1002/adrr.202500180>
- Huzaifa Unjhwala, Luning Bakke, Harry Zhang, Michael Taylor, **Ganesh Arivoli**, Radu Serban, Dan Negrut, "A Physics-Based Continuum Model for Versatile, Scalable, and Fast Terramechanics Simulation," *accepted for publication in Journal of Terramechanics* (2026). <https://arxiv.org/abs/2507.05643>

* Equal contribution